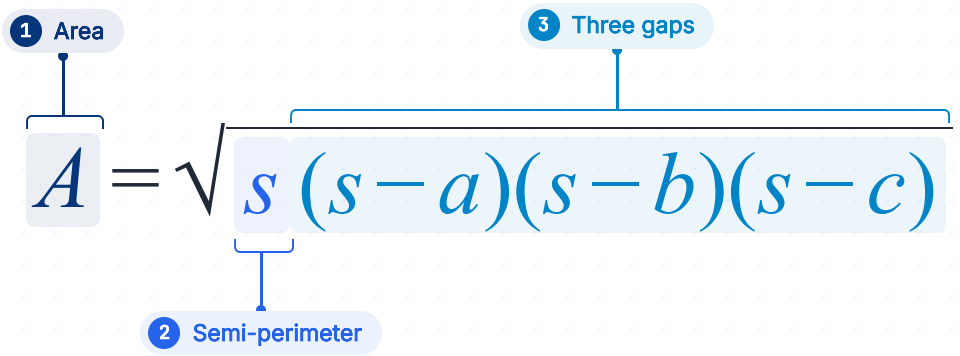
THE FORMULA, PART BY PART

Heron's formula

the area of a triangle from its three sides alone, with no angle needed



1 Area A

No angle and no trigonometric function is needed, but no angle comes out either. If an angle is also wanted, the Law of Cosines is the better start. Example: sides 5, 7, 9 give A = √303.1875 ≈ 17.41.

2 Semi-perimeter $s = (a + b + c)/2$

Add the three sides and halve. Using the full perimeter is the usual slip, and it gives an area far too large. Example: s = (5 + 7 + 9)/2 = 10.5; with s = 21 the result would be about 237.6.

3 Three gaps $s - a,\ s - b,\ s - c$

Multiply s by the three gaps and take the square root. A gap that is zero or negative means the lengths break the triangle inequality and form no triangle. Example: 10.5 − 5 = 5.5, 10.5 − 7 = 3.5, 10.5 − 9 = 1.5, and 10.5 × 5.5 × 3.5 × 1.5 = 303.1875.